# AFFINE SCHEMES AND MORPHISMS FROM THEM ARE SEPARATED

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 4609-4617, following the statement at 4603-4607.

*Allowed standard prerequisites (editorial).*

- Affine diagonals are closed. lemma-diagonal-affines-closed (card L26429 this batch, cited at ~4611). U -> Spec(Z) has closed diagonal.
- Cancellation for separated morphisms. lemma-compose-after-separated (card L26398, cited at ~4614). Separatedness passes to U -> X.

*Source reference.* Original definition 01KK: sources/stacks-schemes/pinned-original-source.tex, source lines 4279–4279. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 01KV: sources/stacks-schemes/pinned-original-source.tex, source lines 4554–4554. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 01KI: sources/stacks-schemes/pinned-original-source.tex, source lines 4239–4239. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *An affine scheme is separated. A morphism from an affine scheme to another scheme is separated.*

*Proof.* Let $U = \mathrm{Spec}(A)$ be an affine scheme. Then $U \to \mathrm{Spec}(\mathbf{Z})$ has closed diagonal by Lemma 01KI. Thus $U$ is separated by Definition 01KK. If $U \to X$ is a morphism of schemes, then we can apply Lemma 01KV to the morphisms $U \to X \to \mathrm{Spec}(\mathbf{Z})$ to conclude that $U \to X$ is separated. □

*Date*: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.